

Project Presentation



Data Analysis for CoffeeKing Expansion

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STEP 1

Preparing for my Proposal



Project Background

- This data analysis project focused on **CoffeeKing** and employed the Yelp Data to provide insights to CoffeeKing expansion regarding location selection, hours of operation and customer perceptions through reviews/ratings in Nashville, TN.
- CoffeeKing was selected for the data analysis because it is a new startup coffee company that seeks to provide a unique and novel experience to their customers in Nashville, TN.
- As part of the Business Goal to appeal to a wide variety of clientele, it is imperative to make data driven and strategic decisions with its limited resources, hence there is a need for a data analysis project for CoffeeKing.

Importing the Data

- I created and connected to the SQL database in the DBeaver Community software.
- After creating the database connection, I created the tables into the database to enable me import the raw data.
- I moved the raw Yelp data CSV files from its storage location on my computer into SQL processing environment with the DBeaver community software.

Cleaning The Data

- As part of the data cleaning efforts I Located null, blank, and NaN values across all columns and removed rows and columns with missing data which lacks a unique ID.
- I filled in missing gaps using statistical measures mean and median, and logical and searched for identical rows or repeating unique identifiers.
- I dropped copies, by keeping the most recent occurrence and converted numbers stored as text into actual numeric formats so I can perform math on them. I converted highly varied date strings into a uniform standard format (YYYY-MM-DD).

Data Exploration

The screenshot displays the DBeaver 26.1.5 interface. The left sidebar shows the database structure for 'localhost 3' (localhost:3306), including a 'project work' database with tables like 'business', 'checkin', 'review', 'tip', and 'user'. The main SQL editor contains the following query:

```
SELECT
COUNT(*) AS total_rows,
(SELECT COUNT(*)
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'business') AS total_columns
FROM business;
```

The 'Results 1' pane shows the query output in a grid format:

	123 total_rows	123 total_columns
1	137	15

A 'Value' window on the right shows the value '137'. The bottom status bar indicates '1 row(s) fetched - 0.144s (0.002s fetch), on 2026-08-23 at 14:22:03'. The bottom toolbar includes options like 'Refresh', 'Save', 'Cancel', 'Export data', and a page size of '200'.

DBEaver 26.1.5 - <localhost 3> Script-1

File Edit Navigate Search SQL Editor Database Window Help

Auto localhost 3 project work Project work ERD... checkin ERD Project-1.erd tip review

Filter connections by name

- localhost localhost:3306
- localhost 3 localhost:3306
 - Databases
 - phpmyadmin
 - project work
 - Tables
 - business 48K
 - checkin 1.5M
 - review 16K
 - tip 16K
 - user 16K
 - Views
 - Indexes
 - Procedures
 - Triggers
 - Events
 - test
 - Users
 - Administer
 - System Info

Files

- Bookmarks
- Dashboards
- Diagrams
- Scripts

```
SELECT
  COUNT(*) AS total_rows,
  (SELECT COUNT(*)
   FROM INFORMATION_SCHEMA.COLUMNS
   WHERE TABLE_NAME = 'checkin') AS total_columns
FROM checkin;
```

Results 1

SELECT COUNT(*) AS total_rows, (SELECT (Enter a SQL expression to filter results (use Ctrl+Space)

	123 total_rows	123 total_columns
1	6,178	4

Value

6178

Refresh Save Cancel Export data 200 1

1 row(s) fetched - 0.017s, on 2026-08-23 at 14:27:54

General CEST en Writable Smart Insert 3:22:59

DBBeaver 26.1.5 - checkin

File Edit Navigate Search SQL Editor Database Window Help

Auto localhost 3 < N/A >

Connections Projects

Filter connections by name

localhost localhost:3306

localhost 3 localhost:3306

Databases

- phpmyadmin
- project work
 - Tables
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Files

Name DataSource

- Bookmarks
- Dashboards
- Diagrams
- Scripts

Properties Data Diagram

Table Name: checkin Engine: InnoDB Collation: utf8mb4_general_ci

Description: Auto Increment: 0 Partitioned

Charset: utf8mb4

	Name	Value
Columns	Row Count	6,115
Constraints	Avg Row Length	259
Foreign Keys	Data Length	1.5M
References	Max data length	0
Triggers	Data free	4M
Indexes	Index length	0
Partitions	Row format	Dynamic
Statistics	Create Time	2026-07-27 22:09:45.0
DDL	Update time	
Virtual	Check time	

Refresh Save ... Revert

General localhost 3 project work checkin CEST en

DBeaver 26.1.5 - <localhost 3> Script-1

File Edit Navigate Search SQL Editor Database Window Help

localhost 3 project work

Filter connections by name

- localhost localhost:3306
- localhost 3 localhost:3306
 - Databases
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Files

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DataSource

SQL

```
(SELECT * FROM business ORDER BY business_id ASC LIMIT 1)
UNION ALL
(SELECT * FROM business ORDER BY business_id DESC LIMIT 1);
```

Results 1

(SELECT * FROM business ORDER BY business_id ASC LIMIT 1) UNION ALL (SELECT * FROM business ORDER BY business_id DESC LIMIT 1);

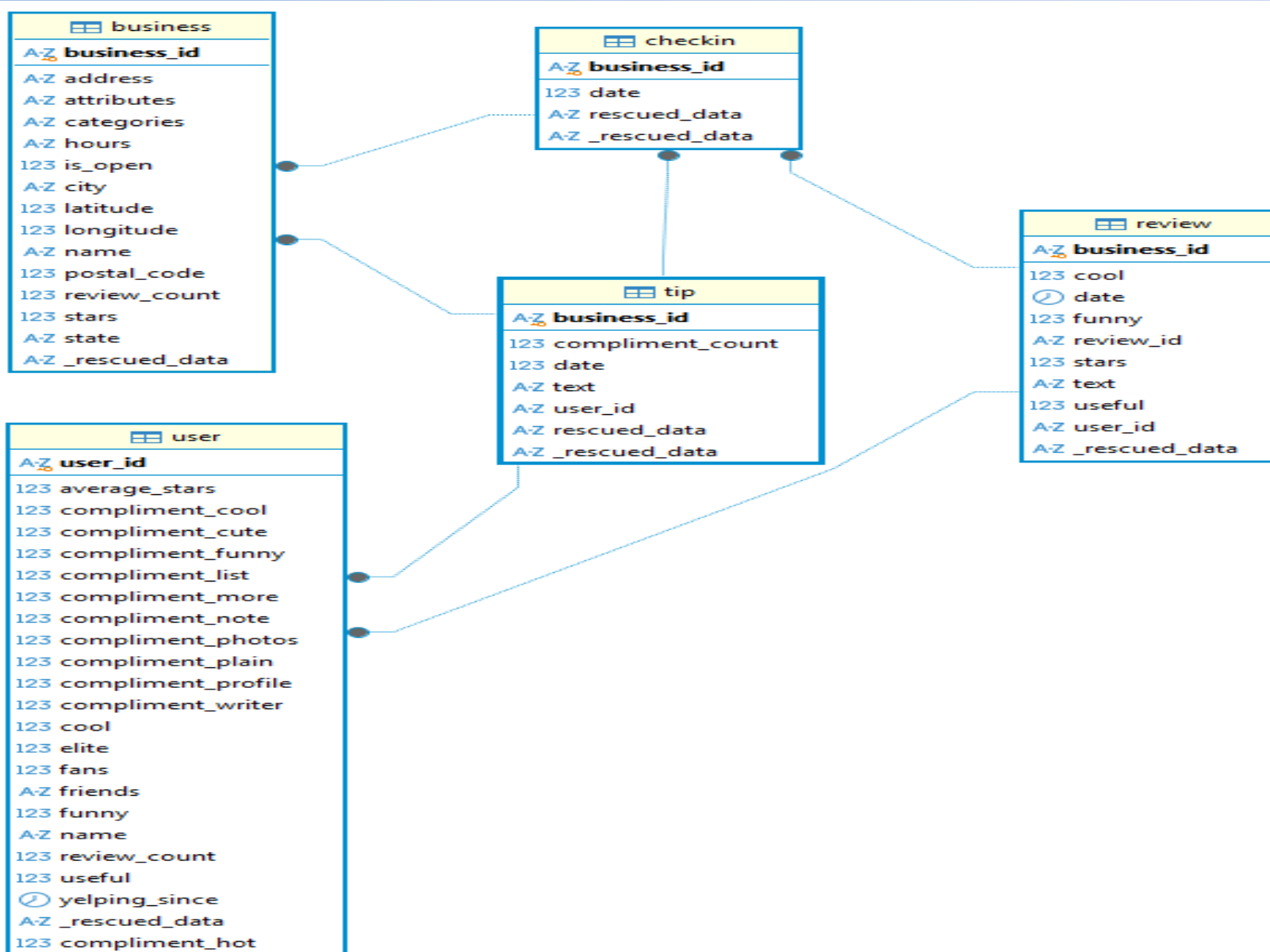
	AZ business_id	AZ address	AZ attributes	AZ categories	AZ hours	123 is_open	AZ city	123 latitude	123 longitude
1	-7oegtOMeB8Cr5gWZUe50A	425 6th Ave S	null	Arts & Entertainment	null	0	Nashville	36	
2	_rLAo3liokiOV2WB_-DyWQ	5055 Broadway Place	null	Hot Dogs, Burgers, Restaurants	null	1	Nashville	36	

Refresh Save Cancel

Export data 200 2 2 row(s) fetched - 0.014s (0.001s fetch), on 2026-08-23 at 14:40:22

General CEST en Writable Smart Insert 4:1:131

Proposed ERD



STEP 2

Developing my Project Proposal



Project Description

- This is a data analysis project proposal for CoffeeKing expansion. The project explored the Yelp Academic Data for Nashville, TN.
- The project provides insights to CoffeeKing regarding location selection, hours of operation and customer perceptions through reviews/ratings in Nashville, TN.
- This project will be significant to CoffeeKing and any other startup coffee making business interested in expanding into the Nashville, TN.
- The target audience of this project will be the coffee businesses and coffee tea customers in Nashville, TN.

Project Questions

1. Which closing hours of operation attract more customers to a coffee shop at Nashville, TN?
 - a. This can show the appropriate closing hours that is convenient for customers.
 - b. It can also reveal the satisfied working hours that appeals to the Labor force at Nashville, TN.
2. Which address areas in Nashville, TN has the most repeated coffee shop checkin and reviews?
 - a. It can show where the most repeated customers come from within Nashville, TN.
 - b. This will give an idea of coffee tea consumption rate in the area.
3. How many customers gave a 5 review stars compared to 1 review star?
 - a. To have a fair idea of the market share to compete for at Nashville, TN.
 - b. This can reveal the satisfaction level of customers towards coffee businesses in Nashville, TN.

Initial Hypotheses

1. Most customers buys from coffee shops that closes by 15hr at Nashville, TN.
 - a. Due to closing hours of most workers.
 - b. Also, most people will like to eat a heavier meal after work.
2. Customers from the 214-11th Ave S has the most most repeated coffee shop checkin and reviews.
 - a. As customers in this area are in the center of the city.
 - b. With the reason that area has more population.
3. More customers gave a 1 review star compared to the 5 review stars at Nashville, TN.
 - a. Mainly because I think most might not be satisfied with the current coffee businesses at Nashville, TN.

Data Analysis Approach

1. I will be looking at the **addresses, stars** and **review counts** columns first to have a fair idea of the frequency rate.
 - a. For location analysis
 - b. To understand the satisfaction level of customers.
2. I will be focusing more on **Frequency Metrics** by paying more attention the hour of operation and review stars to understand how often customers considers the closing hours of coffee shop before visits and also give good review stars.

Final Presentation and Findings

Review of Project Questions

1. Which closing hours of operation attract more customers to a coffee shop at Nashville, TN?
2. Which address areas in Nashville, TN, have the most repeated coffee shop check-ins and reviews?
3. How many customers gave a 5-star review compared to 1 review star?

Review of Initial Hypotheses

1. Most customers buy from coffee shops that close by 15 hr at Nashville, TN.
2. Customers from 214-11th Ave S have the most repeated coffee shop check-ins and reviews.
3. More customers gave a 1-star review compared to the 5-star reviews at Nashville, TN.

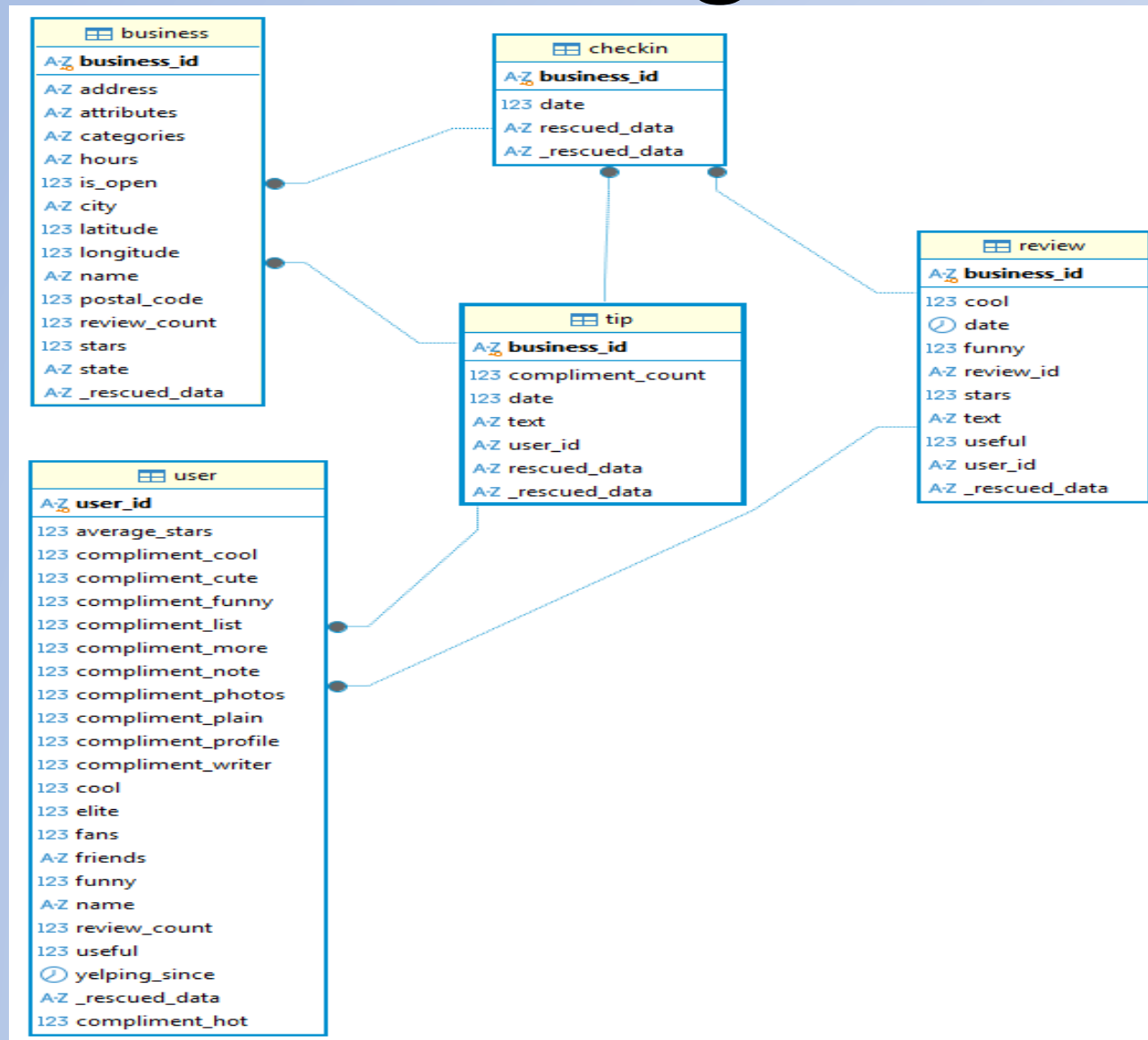
Review of Data Analysis Approach

1. I will be looking at the addresses, review stars, and review counts columns first to have a fair idea of the **frequency rate**.
2. I will be focusing more on **frequency metrics** by paying more attention to the hours of operation and reviewing stars to understand how often customers consider the closing hours of the coffee shop before visits and also give good review stars.

Technical Challenges

- During the investigations and analysis, I encountered some challenges getting some of the addresses that have the most visits to coffeeshops and the most reviews.
- I wanted to know the satisfaction level of the current coffeeshop customers in the area and what most of the customers wrote in their reviews before the star rating. However, I had to use word-scraping in order to get the full text of the reviews written by customers.
- After being able to use word-scraping, I was able to get additional review comments given by customers.

Entity Relationship Diagramm (ERD)



Initial Findings

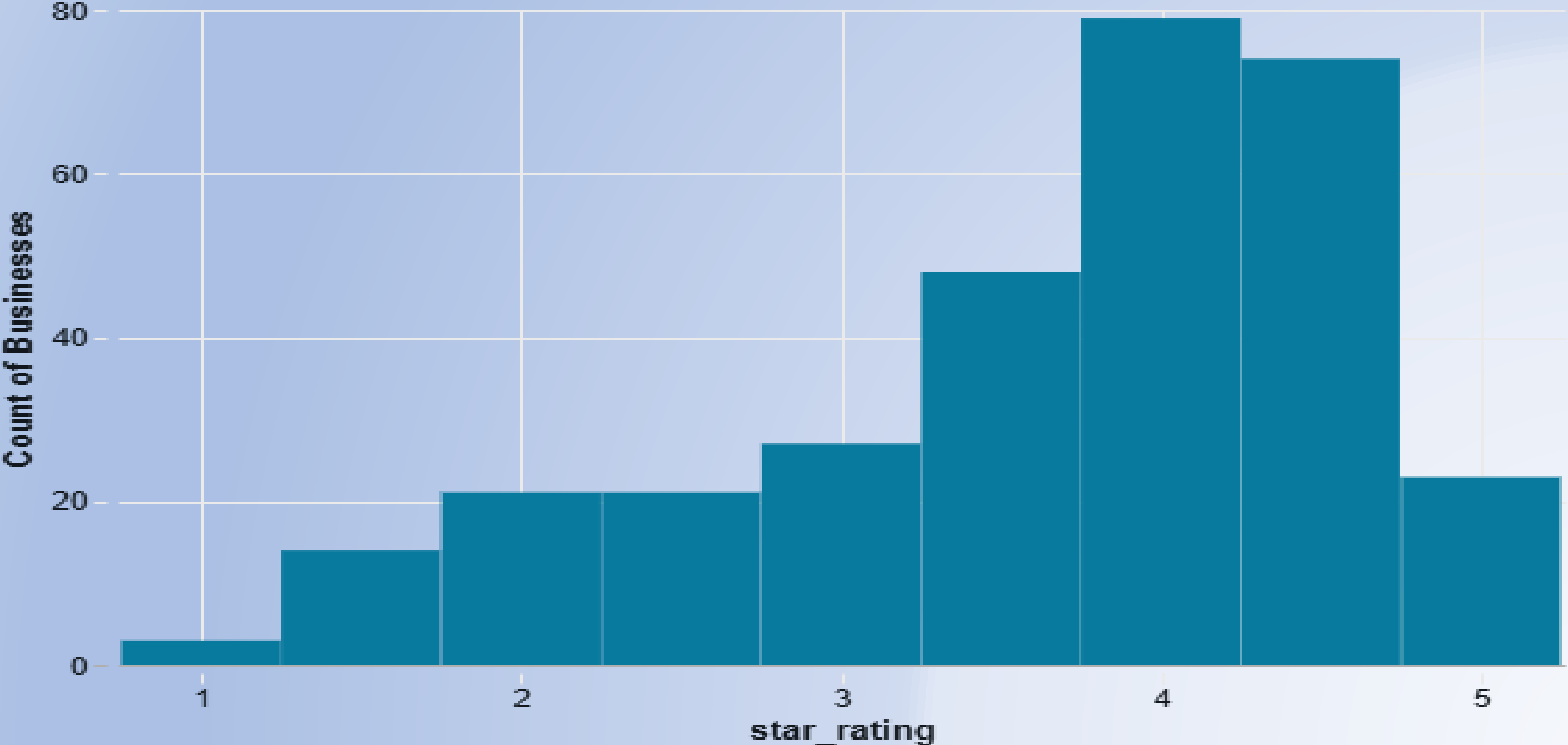
- In my investigation and analysis, I discovered that there is a strong positive correlation between the text reviews and the satisfaction level of coffee and tea customers in Nashville.
- I observed a correlation between the Review Table and the Users table. From these correlations, I learned that some of the text reviews had negative sentiments, and these sentiments affected the star ratings of customers.
- The key terms I discovered in the text analysis are "friendly," "unfriendly," and "fast" staff.

Initial Findings Cont'd

- This implies that coffee and tea customers are more attracted to shops that have friendly staff and offer fast delivery services.
- The results from the analysis show 310 coffee shops in Nashville with ratings ranging from 1 to 5 stars.
- The distribution reveals that 4-star rated coffee shops are most common (79 businesses), followed by 4.5-star (74) and 3.5-star (48) establishments.

Deeper Analysis

Coffee Shops by Star Rating



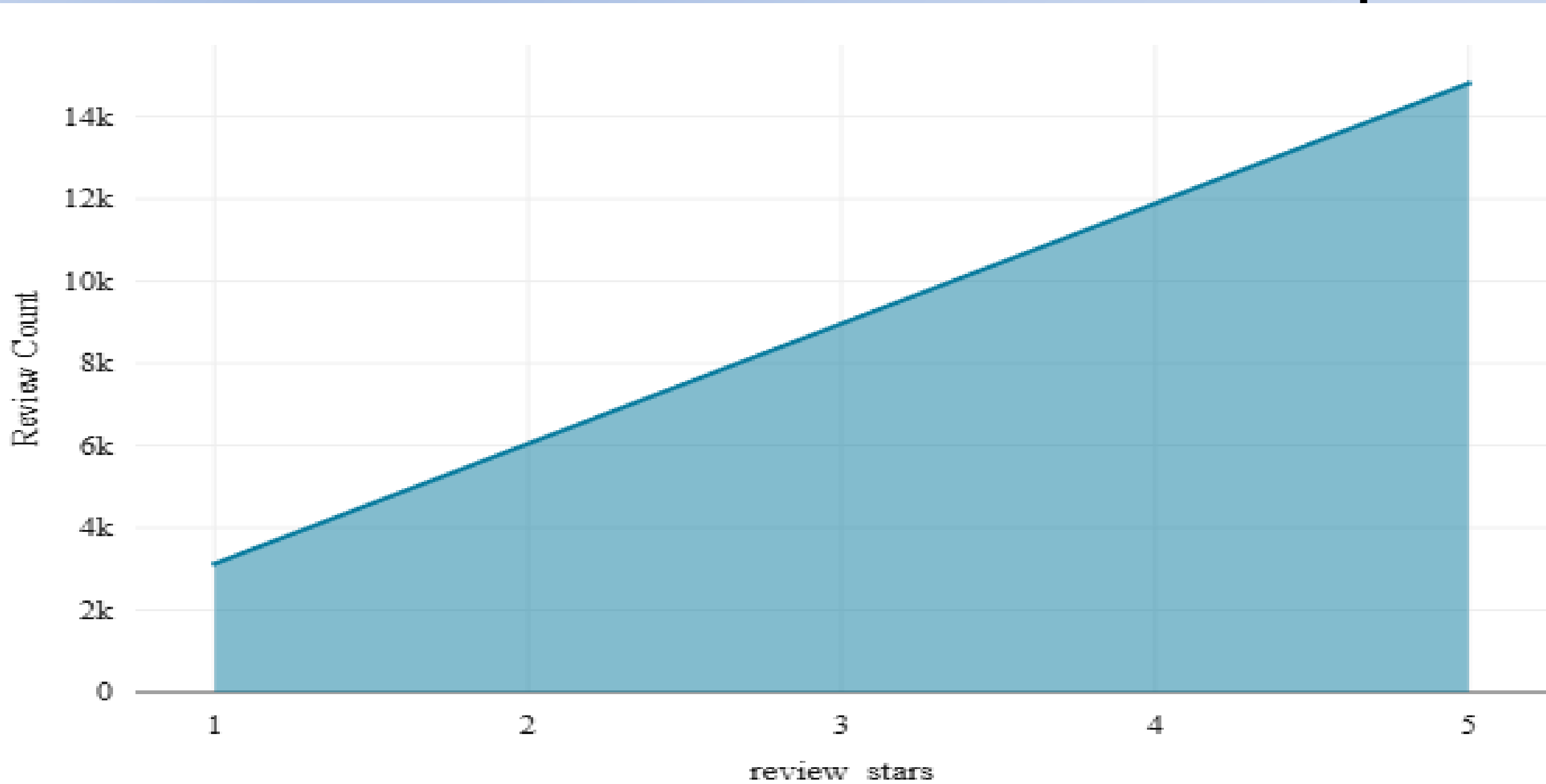
Coffee Shop Addresses by Review Count



Coffee Shops by Closing Hour Comparison



1-Star vs 5-Star Reviews for Coffee Shops



Final Findings (Results of Hypotheses)

1. Most customers buy from coffee shops that close by 15 hr at Nashville, TN.

- a. This happened to be true, as the data clearly shows that early closing at 15 hr is the most common pattern for Nashville coffee shops.
- b. The findings revealed that coffee shops that close at 15 hr have the highest engagement with reviews, as this implies that they serve a significant customer base despite shorter hours.

2. Customers from 214-11th Ave S have the most repeated coffee shop check-ins and reviews.

- a. Yes, customers from 214-11th Ave S have the highest review count with 1,725 reviews for a single coffee shop. This suggests that customers from 214-11th Ave S have the most repeated coffee shop check-ins and reviews.
- b. However, there are some addresses that also revealed a high number of check-ins, which can be considered by CoffeeKing, such as 1812 21st Ave S, which has the second highest with 1,260 reviews, and 15 Hermitage Ave with the third highest of reviews, with 998 reviews.

3. More customers gave a 1-star review compared to the 5-star reviews at Nashville, TN.

- a. This hypothesis tends to be false, as the 5-star reviews for coffee shops seem to be more than the 1-star. The 5-star reviews outnumber 1-star reviews by roughly 4.75x, as 5-star reviews were 14,812 and 1-star reviews were 3,119.
- b. This indicates a strong positive skew in customer sentiment toward coffee shops in the dataset, implying that more customers are satisfied with the current coffee shop businesses. This means that CoffeeKing needs to put in more efforts to win customer loyalty and trust.

Recommendations

- As the customers are satisfied with the current coffee shop businesses in Nashville, CoffeeKing needs to put in more efforts to win customer loyalty and trust.
 1. By intensifying Marketing Campaign efforts via Promotion, advertisements and loyalty bonuses for customers.
 2. By organising regular training for staff to improve customer service efficiency and communication.
 3. By penetrating the Nashville market through competitive pricing by setting prices directly based on what competitors charge.

A top-down view of approximately ten hands of various skin tones reaching up from the bottom and sides to form a circle. The hands are positioned with palms facing upwards. In the center of this circle, the words "THANK YOU" are written in a bold, dark blue, sans-serif font. The text is enclosed within a glowing, light blue, rounded rectangular frame. Surrounding the hands and the central text are various digital and futuristic graphic elements. These include thin, light blue lines, some of which are dashed or dotted, and small, stylized icons resembling triangles and squares. The background is a blurred, light blue and white, suggesting an indoor setting with large windows or a bright, airy space. The overall composition conveys a sense of unity, teamwork, and appreciation.

THANK YOU